

The Legend Challenge

Introduction

This challenge comparatively explores methods for embedding ethical AI directly into models. Using examples from sustainable finance and gender equality, you are requested to demonstrate a novel approach to embedding regulations by using regulation text to generate "champion" positive examples, called "legends," that will inform the LLM model decision-making. You will design and execute an experiment to compare the legend approach to directly including the regulation text in the model training data.

The challenge in detail

1. Request from Dr. Lara Mauri an invitation for your group to our project on the FineTuneDB platform and make yourself familiar with the interface via the tutorials
2. Read the paper "Using Legends to Embed Ethics Into AI-based Decision-Making", available on the course Ariel site
3. Choose one of the regulations on *gender equality* or on *sustainable finance* mentioned in the paper's introduction.
4. Use an LLM of your choice between Grok, Copilot, and ChatGPT to generate five legends, i.e., stories with characters showing concrete compliance with the regulations. Use the same prompt across all LLMs (For example: "Prepare a story involving fictitious characters and showing a concrete example of compliance to the EU Gender Equality Strategy"). See Appendix A for a sample legend.
5. Devise a strategy for converting your legends and the chosen regulation text into JSONL format to match the FineTuneDB input format.
6. Choose one of the free open-pretrained LLMs available on FineTuneDB and tune it using the legend inputs.
7. Use the same open-pretrained LLM and tune it using the regulation text (the total number of lines in the JSONL encoding must be roughly the same for the two models).
8. Consult the GLUE (General Language Understanding Evaluation) benchmark website, which provides comprehensive information about the tasks, datasets, and evaluation metrics used in GLUE¹

1. ¹The original research paper titled "GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding" by Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill,

9. Propose an adaptation of the GLUE benchmark for LLMs to the topic of your choice, gender equality or sustainable finance (guidance in Appendix B)
10. Run your benchmark for the open pretrained LLM and the two versions tuned (i) using legends and (ii) using the regulation text.
11. Conduct an explainability analysis using an appropriate interpretability method (e.g., SHAP, LIME, attention visualization, etc.) applied to a sample of model predictions. Use this analysis to examine how each model justifies its outputs.
12. Prepare a visualization of your results (it is ok to use an LLM for that as well 😊)

Appendix A: Legend generated by CoPilot

Sure, I'd be happy to create a story that illustrates compliance with the Gender Equality Strategy 2020-2025. Here is a fictional narrative:

Title: A New Dawn at Tech Innovators

In the bustling city of Techville, Tech Innovators was a leading company in the technology sector, known for its cutting-edge solutions and innovative products. However, the company had recently decided to take a significant step towards gender equality, inspired by the European Commission's Gender Equality Strategy 2020-2025.

Characters:

- **Alex:** A talented software engineer who had been with Tech Innovators for five years.
- **Jordan:** The newly appointed Chief Diversity Officer, tasked with implementing the gender equality strategy.
- **Taylor:** A project manager who had been advocating for more inclusive policies within the company.
- **Morgan:** A junior developer who had recently joined the company and was eager to see positive changes.

The Story:

Jordan had just joined Tech Innovators and was determined to make a difference. The first step was to conduct a thorough review of the company's policies and practices. Jordan discovered that while Tech Innovators had a diverse workforce, there were significant gaps in gender representation, especially in leadership roles.

Jordan organized a meeting with the senior management team, including Alex, Taylor, and Morgan, to discuss the findings and propose a plan of action. The meeting was held in the company's bright and modern conference room, where everyone gathered around a large table, ready to brainstorm solutions.

"Thank you all for being here," Jordan began. "Our goal is to align Tech Innovators with the Gender Equality Strategy 2020-2025. We need to ensure that our workplace is inclusive and that everyone has equal opportunities to thrive."

Taylor nodded in agreement. "I've been advocating for these changes for a while. It's great to see that we're finally taking concrete steps."

Jordan outlined the key objectives of the strategy: ending gender-based violence, challenging gender stereotypes, closing gender gaps in the labor market, achieving equal participation across different sectors, addressing the gender pay gap, and achieving gender balance in decision-making.

"We need to start by addressing the gender pay gap," Jordan continued. "We'll conduct a comprehensive audit of our salaries to ensure that everyone is paid fairly for their work."

Alex, who had always been supportive of gender equality, added, "We should also focus on mentorship programs. Let's create opportunities for women to advance in their careers and take on leadership roles."

Morgan, the junior developer, chimed in, "I think it's important to challenge gender stereotypes in our workplace culture. We need to promote a culture where everyone feels valued and respected."

Over the next few months, Tech Innovators implemented several initiatives to comply with the Gender Equality Strategy. They introduced mentorship programs, conducted salary audits, and organized workshops to challenge gender stereotypes. The company also launched a campaign to raise awareness about gender-based violence and provided resources for employees who needed support.

The impact was profound. Alex noticed that more women were stepping into leadership roles, and the gender pay gap was gradually closing. Taylor saw a positive shift in workplace culture, where everyone felt more included and valued. Morgan felt proud to be part of a company that was committed to making a difference.

One day, Jordan received an email from the European Commission, congratulating Tech Innovators on their efforts to comply with the Gender Equality Strategy 2020-2025. The email highlighted the company's achievements and encouraged them to continue their work towards a gender-equal society.

Jordan shared the news with the team, and they celebrated their success. "This is just the beginning," Jordan said with a smile. "We're on the right path, and together, we can achieve even more."

This story illustrates how a company can take concrete steps to comply with the Gender Equality Strategy 2020-2025, creating a more inclusive and equitable workplace for all employees.

Appendix B: Adapting GLUE

Adapting the GLUE benchmark for questions about gender equality involves creating a specialized set of tasks and datasets that focus on evaluating a model's understanding and handling of gender equality issues. Here's a step-by-step approach to achieve this:

1. Define the Objectives

The primary goal is to evaluate how well language models understand and process information related to gender equality. This includes:

- Identifying gender biases in text.
- Understanding gender-related policies and their implications.
- Evaluating the sentiment and stance towards gender equality topics.

2. Select or Create Relevant Datasets

You will need datasets that specifically address gender equality issues. These datasets can be sourced from:

- **Policy Documents:** Texts from gender equality strategies, such as the European Commission's Gender Equality Strategy 2020-2025.
- **News Articles:** Articles discussing gender equality, gender-based violence, and related topics.
- **Social Media Posts:** Tweets or posts that discuss gender equality, which can be annotated for sentiment and stance.

3. Design Specific Tasks

Adapt the existing GLUE tasks or create new ones tailored to gender equality. Here are some examples:

- **Text Classification:** Classify texts based on their stance towards gender equality (supportive, neutral, against).
- **Sentiment Analysis:** Determine the sentiment of texts discussing gender equality issues.
- **Bias Detection:** Identify and highlight gender biases in given texts.
- **Reading Comprehension:** Answer questions based on passages from gender equality policy documents.

4. Annotate the Data

Ensure that the datasets are annotated with labels relevant to the tasks. For example:

- **Stance Classification:** Label texts as supportive, neutral, or against gender equality.
- **Bias Detection:** Annotate sentences or phrases that exhibit gender bias.

5. Evaluate and Benchmark

Create a benchmark similar to GLUE, where models are evaluated based on their performance across the different tasks. This can include:

- **Accuracy:** For classification tasks.
- **F1 Score:** For tasks with imbalanced classes.
- **Precision and Recall:** For bias detection tasks.

Example Task: Bias Detection

Dataset: A collection of sentences from various sources. **Task:** Identify sentences that contain gender bias. **Annotation:** Sentences are labeled as biased or unbiased.

Example:

- **Sentence:** "Women are not as good at coding as men."
- **Label:** Biased

Example Task: Stance Classification

Dataset: Social media posts discussing gender equality. **Task:** Classify the stance of each post. **Annotation:** Posts are labeled as supportive, neutral, or against.

Example:

- **Post:** "Gender equality is essential for a fair society."
- **Label:** Supportive

Implementation

1. **Data Collection:** Gather texts from policy documents, news articles, and social media.
2. **Annotation:** Use human annotators to label the data.
3. **Model Training:** Train models on the annotated datasets.
4. **Evaluation:** Use the benchmark to evaluate model performance.

By following these steps, you can adapt the GLUE benchmark to focus on questions about gender equality, providing a robust framework to evaluate and improve language models in this crucial area.